Engineering Mathematics - I · Unit_04 · Unit 4

DRAFT — UE26MA141A · Unit_04 · Unit 4 · Series solution of Besselâ??s differential equation, Bessel functions

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · Series solution of Besselâ??s differential equation, Bessel functions
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

Series solution of Besselâ??s differential equation, Bessel functions

1 Series solution of Besselâ??s differential equation, Bessel functions

Study focus: Series solution of Besselâ??s differential equation, Bessel functions. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000070 · EV0000071

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We are discussing special functions.

Key details

- In today's class, you are going to learn Basel differential equation of order n, series solution of Basel differential equation, and Basel function.
- A linear second order ordinary differential equation in the form x square into y double dash plus x into y dash plus x square minus n square into y is equal to zero.
- Is known as Basel differential equation.
- Here, n is the order of the basal differential equation.
- Let me call this equal equation number 1.
- That is why we call it as series solution of the Basel differential equation.
- And article a0, a1, a2, etc. are constants and a0 is not equal to 0.
- So because we all know that d by dx of x to the power n is equal to n into x to the power n minus 1.
- Therefore, differentiation of x to the power r plus k is r plus k into x to the power r plus k minus 1.
- into x to the power r plus k minus 2.
- Here we have differentiated x to the power r plus k minus 1, which is r plus k minus 1 into x to the power r plus k minus 2.
- We get x square into summation r from 0 to infinity, ar into r plus k into r plus k minus 1.

- into x to the power r plus k minus 2 that is nothing but equal to y double dash this is y double dash then plus x into summation ar into.
- R plus k into x to the power r plus k minus 1.
- This is nothing but equal to y dash, then plus x square minus n square into summation ar into x to the power r plus k.
- This is nothing but equal to y.
- is equal to 0.
- Thus we have substituted the expressions for y dash and y double dash in the x equation x square y double dash plus x y dash plus x square minus n square into y is equal to z, which is a basal differential equation of order n.
- In the second term also I will take this x inside the summation.
- Thus, 1 is x square into summation ar x power r plus k.
- That is nothing but equal to summation r from 0 to infinity, ar into x to the power r plus k plus 2, then minus n square into summation r from 0 to infinity ar into x to the power r plus k is equal to 0.

Example or application from the lecture

- Let us call it as equation number 2, where k is a positive real number.
- Let us call it as equation number four.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

Core explanation · 05:00–09:59

That implies summation r from 0 to infinity ar into r plus k into r plus k minus 1 plus 1 into x to the power r plus k.

Key details

- We have considered the first two terms together here.
- Then minus n square into summation r from zero to infinity ar plus k plus summation r from zero to infinity ar x to the power r plus k plus two is equal to zero.
- Here we can cancel these two terms.
- Summation R from zero to infinity into x to the power r plus k plus two is equal to zero.
- Summation R from 0 to infinity R plus K whole square minus N square into AR into X to the power R plus K.
- Summation R from 0 to infinity is common.
- A R is common factor, x power r plus k is also common, so we can write it as r plus k whole square minus n square.
- Plus summation r from 0 to infinity a r into x to the power r plus k plus 2 is equal to 0.
- That implies summation r from 0 to infinity ar into x r plus k whole square minus n square into x to the power r plus k plus.
- Summation R is equal to 2 to infinity AR minus 2 into x to the power r plus k is equal to 0.
- See we obtain this term by replacing r by r minus 2 in the last term here.

- And here the summation starts from r minus 2 is equal to 0 to infinity, that is nothing but r is equal to 2 to infinity.
- In one summation R starts from 0 and in another R starts from 2.
- So, in the first summation, I will write two terms separately, that is, when r is equal to 0 and when r is equal to 1.
- That implies a0 into k square minus n square into x power k plus.
- A1 into K plus one whole square minus N square into X power K plus one plus summation R from two to infinity A R into R plus K whole square minus N square into X power R plus K plus.
- Summation R from 2 to infinity AR minus 2 into x power r plus k is equal to 0.
- After combining these two summations.
- We get it as summation r from two to infinity AR into R plus K whole square minus N square plus AR minus two whole into x power r plus k is equal to zero.
- Coefficient of x power k is a0 into k square minus n square.
- That implies k square minus n square is equal to 0 because we already know that k0 is not equal to 0.
- This implies a1 is equal to 0.
- So why we got this is because if a1 is not equal to 0, we get k plus 1 whole square minus n square is equal to 0.
- That means k plus 1 whole square is equal to n square.

Example or application from the lecture

- We consider it as a common summation.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

Core explanation · 10:00–14:59

That means k plus 1 is equal to plus or minus n, which is not correct because already we have proved now that k is equal to plus or minus n.

Key details

- Therefore, this term cannot be equal to 0.
- By equating the higher power of x to 0, we get ar is equal to some minus ar minus 2 divided by r plus k whole square minus n square.
- Call it as equation number five.
- First case is when k is equal to n.
- Put k is equal to n in equation number 5, we get ar is equal to minus ar minus 2 divided by r square plus 2 n r.
- Because we know that AR is equal to minus A R minus 2 divided by R plus K whole square minus n square by equation number 5.
- When you put k is equal to n in the denominator you get r plus n the whole square minus n square.
- And R plus N the whole square is R square plus 2 N R plus N square minus N square.

- Call it as equation number 6.
- A0 divided by 2 square plus 4n.
- Because when r is equal to 2, a r minus 2 becomes a 2 minus 2, which is a0.
- And in the denominator when you put r is equal to 2, r square becomes 2 square, 2n r becomes 4n.
- Which is equal to minus of a0 divided by 2 square into n plus 1.
- Therefore, your a2 is equal to minus a0 divided by 2 square into n plus 1.
- Call it as equation number 7.
- Now a3 is equal to minus a 1 divided by t square plus 6n, which is equal to 0 because a1 is equal to 0.
- This is got by ar is equal to minus ar minus 2 divided by r square plus 2 n r by putting r is equal to 3.
- In the numerator when R is 3, you get A3 minus 2, which is A1, which is there in the numerator here, and in the denominator 3 square you get and plus 2 Nr becomes a 6n.
- R is 4, so it becomes 4 square plus 2n r is equal to a n.
- Which is equal to minus 1 divided by 8 into m plus 2 into a2.
- We are substituting for a2 here because we know a2 is equal to minus a0 divided by 2 square into m plus 1.
- Thus we have a4 is equal to a0 divided by 32 into n plus 1 into n plus 2.
- which is equal to zero because we know that a3 is equal to zero.
- We can continue like this.
- Y is equal to a0 x power k plus a1 x power k plus one plus a2 x power k plus two plus a three x power k plus three plus a4 x power k plus four and so on.
- because we have assumed a y as a series solution that is y is equal to summation r from 0 to infinity ar into x to the power k plus r.
- We call it as Y1 because we are just discussing case 1 where your K is equal to plus N.
- We had two cases K is plus or minus N.
- So for the first case K is equal to N we call the solution as Y1.

Example or application from the lecture

- Let us substitute r is equal to 2, 3, 4, etc. in equation number 6, we get when you put r is equal to 2, we get a2, a2 is equal to minus.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 15:00–19:59

So y1 is equal to a0 into x power n plus 0 into x power n plus 1 because we know that a1 is equal to 0 and k is equal to n here.

Key details

- Then plus a2 x power k plus 2.
- Next term is a3x power k plus 3.
- A3 is equal to 0.

- Then a4 x to the power k plus 4.
- Substitute for a4, a4 is a0 divided by 32 into n plus 1 into n plus 2 into x to the power n plus 4 plus other terms.
- That is y1 is equal to a naught into x power n minus 1 divided by 4 into n plus 1 into x power n plus 2 plus 1 divided by 32 into n plus 1 into m plus 2 into x to the power n plus 4 plus other terms.
- Remaining terms I write as it is.
- Minus 1 divided by 4 into n plus 1 into 2 power n into gamma of n plus 1 into x power n plus 2 plus.
- 1 by 32 into n plus 1 into n plus 2 into 2 power n into gamma of n plus 1 into x to the power n plus 4 plus other terms.
- Which is nothing but equal to gamma of n plus 2.
- And then you have n plus 2 into gamma of n plus 2.
- What is n plus 2 into gamma of n plus 2?
- So therefore y1 is equal to 1 divided by 2 power n into gamma of n plus 1 into x to the power n.
- Minus 1 divided by 2 square into 2 power n into gamma of n plus 2 into x power n plus 2.
- Plus 1 divided by 2 into 2 to the power 4.
- Because we had 32 in the denominator, 32 can be written as 2 to the power 5, it can be written as 2 into 2 to the power 4.
- into 2 power n into gamma of n plus 3 into x to the power n plus 4.
- Plus other terms.
- Plus this 1 by 2 we keep it as it is and 2 power 4 into 2 power n can be written as 2 to the power n plus 4 into gamma of n plus 3 into x to the power n plus 4 plus other terms.
- That is equal to C first term, so we can write it as minus 1 power 0.
- Minus 1 power 0 is 1.

Example or application from the lecture

- To standardize the result, let us take a0 is equal to 1 divided by 2 power n into gamma of n plus 1.
- And consider x power n divided by 2 power n, which can be written as x by 2 whole to the power n.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

Core explanation · 20:00–24:59

Then plus, since there is a negative sign here, we can write it as minus 1 whole to the power 1 divided by gamma of n plus 2 into x power n plus 2 divided by 2 to the power n plus 2, which is x by 2 whole to the power n plus 2.

Key details

- Next term is positive, so you can write it as 1 by 2 into minus 1 whole square in divided by gamma of n plus 3 into x power n plus 4 divided by 2 to the power n plus 4, which is x by 2

whole to the power n plus 4 plus other terms.

- That is equal to minus 1 power 0 divided by 0 factorial.
- 0 factorial is 1.
- Into gamma of n plus 1 into x by 2 whole to whole to the power n can be written as x by 2 whole to the power n plus 2 into 0.
- Plus minus 1 power n divided by 1 factorial, 1 factorial is 1, into gamma of n plus 2 into x by 2 whole to the power n plus 2 can be written as x by two whole to the power n plus 2 into 1.
- Plus, 1 by 2 can be written as 1 by 2 factorial and minus 1 whole square divided by gamma of n plus 3 into x by 2 whole to the power n plus 2 into 2.
- That is nothing but x by 2 whole to the power n plus 4 plus other terms.
- We can write it in the summation form as y1 is equal to summation r from 0 to infinity minus 1 to the power r into x by 2 whole to the power n plus 2 r into 1 divided by gamma of n plus r plus 1.
- into r factorial.
- This is because when r is equal to 0, you get minus 1 power 0 x by 2 whole to the power n plus 2 into 0 into 1 divided by gamma of n plus 1 into.
- One factorial, zero factorial.
- That is what the first term here.
- When you put r is equal to one, you get minus one to the power one into x by two whole to the power n plus two into one into one divided by gamma of n plus two into one factorial.
- When you put r is equal to two,.
- You get minus one whole square into x by two whole to the power n plus two into two into one divided by gamma of n plus two plus one that is gamma of n plus three into two factorial that is what the third term here.
- Call it as equation number nine.
- Equation number nine is called power series solution of basal differential equation.
- It is called Basel function of first order n.
- and it is denoted by jn of x.
- Thus jn of x is equal to summation r from 0 to infinity minus 1 power r divided by r factorial into gamma of n plus r plus 1 into x by 2 whole to the power n plus 2 r.
- Remember this expression, it's an important expression.
- Put k is equal to minus n in equation number five.
- Then we get j minus n of x is equal to summation r from 0 to infinity minus 1 power r x by 2 whole to the power minus n plus 2 r into 1 divided by gamma of minus n plus r plus 1 into r factorial.
- Basically, Paz's differential equation of order n is a differential equation of order 2.
- Moreover, we can.
- have a condition that n is a non integer here because we know that for a differential equation of order two, if y one and y two are two linearly independent solutions,.
- Then C1 by one plus c two by two that is the linear combination of those two linearly independent solutions is also a solution of the differential equation.
- Here jn and j minus c n are two linearly independent solutions only if n is not an integer.

Example or application from the lecture

- Thus the complete solution of Basel differential equation of order n is given by y is equal to c1 into j n of x plus c2 into j minus n of x, where c1 and c two are arbitrary constants.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

Core explanation · 25:00–29:59

If n is an integer, jn and j minus n are not linearly independent.

Key details

- You can prove this by taking the expression for jn of x.
- Just replace n by minus n.
- And hence you can prove that j minus n of x is minus 1 to the power n into jn of x.
- That shows that jn and j minus n are not linearly independent when n is an integer.
- Thus, if n is non-integer, then we have y is equal to c1 j n of x plus c2 j minus c n of x as the complete solution of the Bessel's differential equation.
- You can just try this to prove that jn and j minus n are linearly dependent when n is an integer.
- It is interesting, just take it as an assignment and try it on your own.
- With this, we will end this session.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Series solution of Besselâ??s differential equation, Bessel functions transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/010_Series_solution_of_Besselâ_s_differential_equation_Bessel_functions__0b6c9f8e6bffd10d.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Series solution of Besselâ??s differential equation, Bessel functions transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/010_Series_solution_of_Besselâ_s_differential_equation_Bessel_functions__0b6c9f8e6bffd10d.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Series solution of Besselâ??s differential equation, Bessel functions transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/010_Series_solution_of_Besselâ_s_differential_equation_Bessel_functions__0b6c9f8e6bffd10d.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9989; only the JSON segmentation supplies the displayed timestamps.
Series solution of Besselâ??s differential equation, Bessel functions transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/010_Series_solution_of_Besselâ_s_differential_equation_Bessel_functions__0b6c9f8e6bffd10d.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
2. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
3. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
4. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00022	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare...
SRC00023	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00024	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare...
SRC00025	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare...

C Revision and retrieval practice